

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

Packet capture analysis showed repeated UDP traffic directed at port 53 returning an unreachable error. Specifically, the ICMP echo reply returned the message “udp port 53 unreachable, length 254,” confirming that traffic to port 53 was being rejected.

Port 53 is the standard port used by the Domain Name System (DNS) to resolve domain names into IP addresses. Because this port was unreachable, the most likely root cause of the disruption is a failure in DNS resolution, meaning client requests to translate domain names were not completing

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: 13;24:32.192571

Multiple customers and clients reported that they were not able to access the company’s website.

The IT team used a network traffic analyzer to inspect packet-level activity and identify the source of the error.

The analysis traced the issue to port 53, which is used by the DNS server to translate domain names into IP addresses. Repeated “port unreachable” responses on this port indicated that DNS requests were not being processed.

A network misconfiguration or firewall rule was likely blocking or interfering with DNS traffic on port 53, preventing successful name resolution.

Conclusion: Packet capture analysis showed repeated UDP traffic directed at port 53 returning an unreachable error. Specifically, the ICMP echo reply returned the message “udp port 53 unreachable, length 254,” confirming that traffic to port 53 was being rejected.

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